

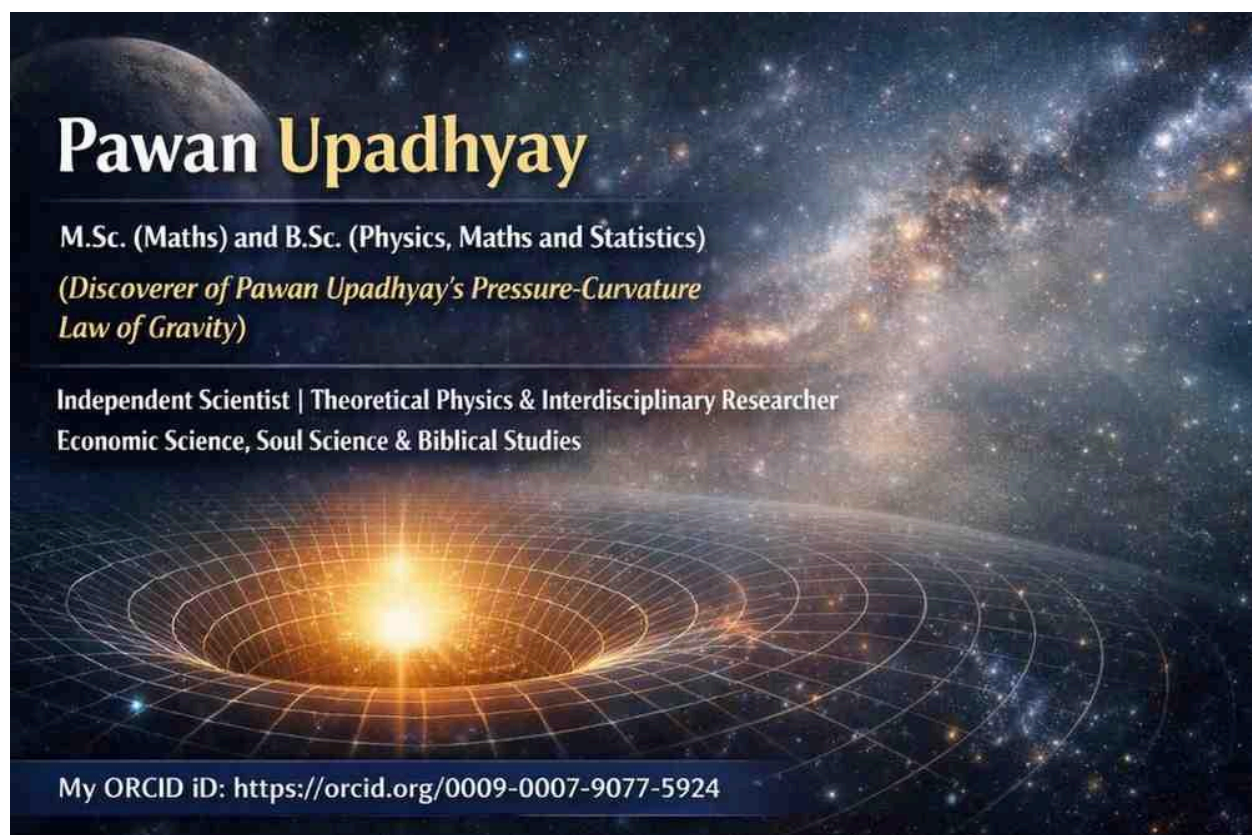
Gravitational Pressure Field, Pressure Gradient Field, and Gravitational Field: A Unified PPC Framework

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Abstract

This paper presents a unified formulation of gravity within the Pressure Curvature (PPC) Law of Gravity by establishing precise relationships among the gravitational pressure field (P_g), its spatial gradient (∇P_g), and the gravitational field (g). We demonstrate that gravitational

acceleration arises from the density-normalized gradient of the pressure field, thereby linking scalar pressure, vector force density, and observable acceleration into a single consistent framework. The formulation is shown to recover Newtonian gravity and Keplerian dynamics in the weak-field limit.

1. Introduction

In classical physics, gravity is described by the gravitational field:

$$\mathbf{g} = -\nabla \phi$$

where ϕ is the gravitational potential. In the PPC framework, gravity is interpreted as a pressure-driven phenomenon characterized by a scalar pressure field P_g .

This work establishes a direct connection between:

- Gravitational pressure field (P_g)
 - Pressure gradient field (∇P_g)
 - Gravitational field ($\mathbf{g}$)
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2. Fundamental PPC Relation

The PPC law is expressed in force-density form:

$$\rho \mathbf{a} = -\nabla P_g$$

Since gravitational field is acceleration per unit mass:

$$\mathbf{g} = \mathbf{a}$$

we obtain:

$$\mathbf{g} = - (1/\rho) \nabla P_g$$

3. Relation to Gravitational Potential

In classical form:

$$\mathbf{g} = -\nabla \phi$$

Comparing with PPC expression:

$$-\nabla \varphi = - (1/\rho) \nabla P_g$$

This implies:

$$\varphi = P_g / \rho$$

Thus, gravitational potential is identified as pressure per unit density.

4. Unified Field Relations

The three fundamental quantities are related as follows:

1. Pressure field: P_g
2. Pressure gradient field: $\nabla P_g = -\rho g$
3. Gravitational field: $g = - (1/\rho) \nabla P_g$

These relations form a closed and consistent system.

5. Physical Interpretation

- P_g represents the scalar distribution of gravitational pressure in spacetime
- ∇P_g represents force density acting on matter
- g represents observable acceleration

Thus:

Pressure field $\rightarrow$ Gradient $\rightarrow$ Force density $\rightarrow$ Acceleration

6. Special Cases

6.1 Constant Pressure Field

If $P_g = \text{constant}$:

$$\nabla P_g = 0 \Rightarrow g = 0$$

No gravitational effects occur.

6.2 Strong Gradient Field

If ∇P_g is large:

g is large

Strong gravitational effects arise.

7. Weak-Field Limit

For a point mass M :

$$\phi = -GM/r$$

Using $\phi = P_g/\rho$:

$$P_g = -\rho GM/r$$

Taking gradient:

$$g = GM/r^2$$

Thus, the PPC formulation reproduces Newtonian gravity.

8. Circular Motion

For circular motion:

$$a = v^2 / r$$

Using PPC relation:

$$\rho v^2 / r = -\nabla P_g$$

This shows that centripetal force density is generated by pressure gradients.

9. Keplerian Consistency

From the above:

$$v^2 = GM/r$$

$$T^2 \propto r^3$$

Thus, Kepler's laws follow naturally from pressure-based dynamics.

10. Conclusion

The PPC framework establishes a direct and consistent connection between gravitational pressure field, its gradient, and the gravitational field. Gravity is interpreted as a manifestation of pressure gradients rather than a fundamental force. This unified formulation reproduces known physical laws while providing a deeper physical interpretation based on pressure.

Keywords

PPC Gravity, Gravitational Pressure, Pressure Gradient, Gravitational Field, Potential, Force Density, Kepler's Laws

Important Point:

Kepler's laws emerge when gravity follows an inverse-square law, which occurs in the weak-field limit of both Newtonian gravity and General Relativity.

In other words

Kepler's laws are valid in the weak-field (Newtonian) limit.

In the weak-field limit, where spacetime curvature is small and gravitational interactions follow an inverse-square law, Kepler's laws of planetary motion are recovered. Within the PPC framework, this corresponds to a regime in which the gravitational pressure gradient behaves as $\nabla P_g \propto 1/r^2$, leading to classical orbital dynamics.

The relation $F = \rho g$ is valid as a force-density expression in the weak-field (Newtonian) limit, where gravitational acceleration is well defined and spacetime curvature is negligible. In the PPC framework, this relation is consistent with $F = -\nabla P_g$, linking gravitational field to pressure gradients. However, in strong-field regimes, a covariant formulation is required.

Near the Neutron Star, the Gravitational Pressure Field is very large and the Pressure gradient is very large too.

- $P_g \rightarrow$ measures **intensity of gravitational pressure field**
- $\nabla P_g \rightarrow$ drives **dynamics (force / curvature effects)**

So:

👉 High P_g + high $\nabla P_g \Rightarrow$ **strong gravity regime**
